TEXORA · LECTURE SHEET

Process Control in Simplex Machine

The simplex (roving frame) is controlled on roving weight — because its error passes straight into yarn count — and on roving tension, which decides whether the roving stretches between drafting and the bobbin. This sheet covers the roving weight test and draft correction, twist control, and package build.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 7 · SPECIALIZATION — YARN MANUFACTURING

1. Roving weight control

Roving is tested by weighing a measured length wound from the bobbin (commonly 15 m or a full lea-length equivalent) and converting to tex or hank. Because ring-frame draft is fixed by change wheels, any roving weight error appears in the yarn, multiplied by nothing and corrected by nothing: roving +1% heavy gives yarn 1% coarse unless the ring draft is changed. The simplex is therefore held to a tight band, typically within about ±1% of nominal hank.

WORKED EXAMPLE — ROVING WEIGHT TEST AND DRAFT CORRECTION

Nominal roving = 1.20 Ne = 492.1 tex. A 15 m length from a bobbin weighs 7.45 g. Draft in use = 7.8.

Tested weight = 7.45/15 × 1000 = **496.7 tex** = Ne 590.5/496.7 = 1.189.

Deviation = (496.7 − 492.1)/492.1 × 100 = **+0.93%** (heavy).

Correction: delivered weight falls as draft rises, so new draft = 7.8 × (496.7/492.1) = **7.87**.

After the change, retest from bobbins built afterwards. Also check the feed: if drawframe sliver went heavy by the same 0.93%, the correction belongs at the drawframe (Topic 5), and the simplex draft should stay — correcting at both places creates a double error.

2. Roving twist control

Roving needs only enough twist to survive unwinding and drafting without stretching — for the 1.20 Ne roving of this course, TM 1.1 gives TPI = 1.1 × √1.20 = 1.21 (YM-3207 Topic 5). Too little twist: roving breaks and stretches in the ring creel, producing thin places that look like drafting faults. Too much: the ring frame's drafting system cannot open the roving cleanly, producing hard ends, thick

places and roller slip. Twist is verified by the untwist method on samples from several spindles, and twist CV across the frame exposes worn flyers and uneven spindle drives.

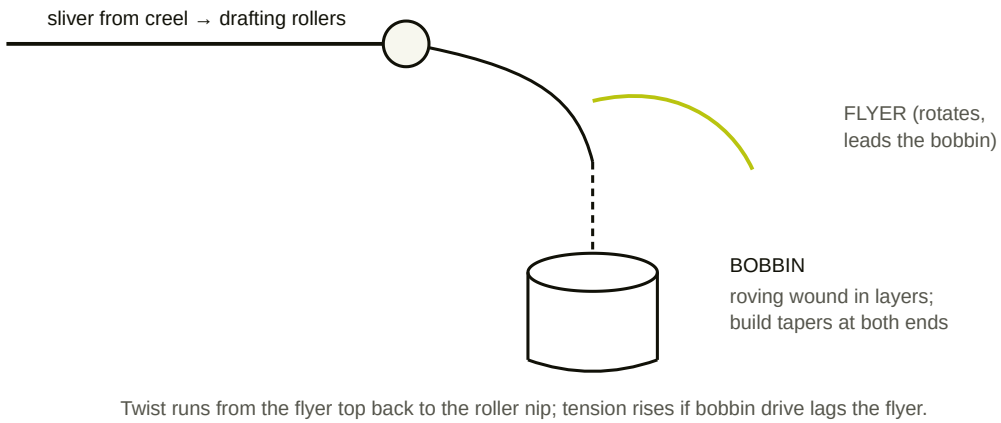


Figure 1 — Roving path in the simplex: drafted strand passes through the hollow flyer leg and is wound on the bobbin. The flyer–bobbin speed difference winds the package; its setting controls tension and build.

3. Tension and package build

The bobbin is driven slightly faster or slower than the flyer as the package grows, so that the roving winds on with constant, gentle tension (the builder motion also traverses the winding point to form the tapered package). Control observations: roving that sags between front roller and flyer guide is under-tensioned and will stretch at the ring frame; roving wound drum-hard signals over-tension — it has already been stretched thinner than its tested weight, which is why roving weight is always tested from the bobbin, not calculated from draft alone. Package faults — ridges, soft ends, collapsed layers — are builder-motion and cone-drum faults, logged per spindle position so repeat offenders can be found.

4. Norms and schedule

CONTROL	TYPICAL NORM	FIRST ACTION ON FAILURE
Roving weight vs nominal	Within ±1%	Draft correction (worked above); check feed sliver first
Roving weight CV (between bobbins)	≲ 1.5%	Find outlier spindle positions; check drafting rollers and skewers
Roving twist	On TM target ±5%	Verify change pinion; untwist test across frame
Package build	Firm, even taper, no ridges	Builder motion and cone-drum adjustment

Breaks at simplex	Low and stable trend	Separate drafting breaks (process) from winding breaks (tension)

Glossary

Simplex — the roving frame; it drafts sliver to roving and winds it on bobbins. **Flyer** — the rotating U-shaped arm that guides roving onto the bobbin and inserts twist. **Builder motion** — the mechanism traversing the winding point to build the tapered package. **Cone drum** — the friction drum whose varying drive compensates bobbin diameter growth. **Stretch** — unintended drafting of roving under tension between roller and bobbin, or in the ring creel. **Hard ends** — ring-frame faults where over-twisted roving refuses to draft. **Skewer** — the rod carrying the roving bobbin in the ring-frame creel.

Key points

• Roving weight error passes unchanged into yarn count — hence the ±1% band and bobbin-based testing.
• Test: 15 m weighing 7.45 g = 496.7 tex, +0.93% heavy; correction draft 7.8 → 7.87, then retest.
• Check the feed sliver before correcting: never let two departments correct the same error.
• Twist on target (TM 1.1 → 1.21 TPI for 1.20 Ne): too little stretches, too much refuses to draft.
• Tension faults falsify the weight test itself — roving is therefore tested as it comes off the bobbin.

Practice

1. A 15 m roving length weighs 7.20 g. Nominal is 492.1 tex. Calculate the tested tex, the deviation %, and the corrected draft if 7.8 is in use.
2. Roving twist tests read 1.05–1.38 TPI across a frame whose target is 1.21 TPI. What does the spread, rather than the mean, tell you, and what do you inspect?
3. Yarn from one ring frame shows intermittent thin places, yet its roving tests on weight. Give the simplex-side explanation you would check first.

Answer key

1. Tested = 7.20/15 × 1000 = **480.0 tex**. Deviation = (480.0 − 492.1)/492.1 × 100 = **−2.46%** (light). New draft = 7.8 × (480.0/492.1) = **7.61** (draft falls to raise the weight).
2. The mean may be near target while the spread (about ±14%) shows spindle-to-spindle twist variation: worn or dirty flyers, uneven spindle tapes/drives, or inconsistent presser behaviour. Inspect flyers and drives position by position — a single twist change pinion cannot fix a spread.

3. Roving stretching in the ring creel — from under-twisted roving or excessive winding tension at the simplex (drum-hard bobbins). Check twist level and bobbin hardness on the suspect simplex; the roving leaves on-weight but is pulled thin before it reaches the drafting rollers.
